

Workshop Market Analysis — Accelerated Execution Plan

Confirmed Pilot Scope

Cities

- Bengaluru
- Hyderabad

Initial workshop categories

- Cooking and baking
- Art and painting
- Pottery and ceramics
- STEM, coding, and robotics

Initial product types

- Individual workshops
- Camps
- Birthday and private events
- Recurring classes and memberships, where available

Formats

- Dedicated studios
- Home-based workshops
- Rented venues
- Mobile workshops
- Online workshops
- Hybrid workshops

The pilot will validate the data sources, collection methods, database design, metrics, and analysis process before expanding to Mumbai, Delhi NCR, Pune, and Chennai.

Roles During the Project

ChatGPT

ChatGPT will function as the:

- Project planner

- Business analyst
- Research reviewer
- Metric designer
- Data-quality reviewer
- Statistical-analysis assistant
- Insight validator
- Dashboard reviewer
- Documentation and presentation assistant

Codex

Codex will function as the:

- Repository coding agent
- Data-pipeline developer
- Web-collection developer
- PostgreSQL and dbt developer
- Testing assistant
- Notebook developer
- SQL and Python assistant
- Power BI support-code developer

Your role

You will:

- Run and review the code
- Understand the architecture
- Approve collection sources
- Inspect sample data
- Validate business assumptions
- Test PostgreSQL and Power BI connections
- Maintain Git commits
- Explain the project in your own words

The working principle will be:

AI designs and implements quickly; you review, understand, test, and make the final decision.

Required Architecture Updates

Before building collectors, Codex should update the repository documentation with these changes:

1. Replace **Pune** with **Hyderabad** in the pilot scope.
2. Add a `powerbi/` directory for Power BI project files and documentation.

3. Add a `docs/project_log.md` file to record daily progress.
 4. Add a `docs/decision_log.md` file to record architectural and analytical decisions.
 5. Add collection-run metadata, including:
 6. Source
 7. Start and completion time
 8. Number of records discovered
 9. Number successfully collected
 10. Failed URLs
 11. Collection method
 12. Code version
 13. Add manual-validation support for:
 14. Duplicate checks
 15. Category checks
 16. Sold-out-status checks
 17. Organizer matching
 18. Preserve separate values for:
 19. Sold out
 20. Registration closed
 21. Cancelled
 22. Unavailable
 23. Unknown
 24. Do not add Airflow, cloud infrastructure, or complex orchestration during the pilot.
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Accelerated 25-Day Project Plan

This is an aggressive but practical plan for completing the pilot, expanding the analysis, and producing a portfolio-ready output.

Day 1 — Repository and Architecture Review

- Inspect the generated folder structure.
- Compare it with the proposed architecture.
- Update Bengaluru and Hyderabad as pilot cities.
- Create project, decision, and progress logs.
- Confirm Python, PostgreSQL, dbt, Jupyter, and Power BI folders.
- Run the repository's existing tests.

Output

- Approved repository structure
 - Updated architecture document
 - Working project log
 - Clean Git commit
-

Day 2 — Business Taxonomy

Define and store:

- Workshop categories
- Subcategories
- Age groups
- Event types
- Delivery formats
- Operating models
- Price tiers
- Event-status definitions

Output

- `category_taxonomy.csv`
 - `age_group_taxonomy.csv`
 - `format_taxonomy.csv`
 - `price_tier_rules.md`
-

Day 3 — Source Register

Identify potential sources for Bengaluru and Hyderabad.

For each source, document:

- Available fields
- Historical coverage
- Collection method
- Access restrictions
- Reliability
- Update frequency
- City coverage
- Priority

Output

- Source register
 - Source-compliance checklist
 - Shortlisted pilot sources
-

Day 4 — PostgreSQL Environment

- Configure Docker Compose.
- Start PostgreSQL.
- Create database schemas:
 - `raw`
 - `staging`
 - `core`
 - `analytics`
 - `powerbi`
 - `quality`
- Test Python and SQL connections.
- Store credentials securely in `.env`.

Output

- Running PostgreSQL database
 - Successful connection test
 - Schema initialization scripts
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Day 5 — First Collector

Build one simple collector for the most accessible source.

It should:

- Collect a small Bengaluru and Hyderabad sample.
- Save raw responses unchanged.
- Record collection metadata.
- Handle retries and failures.
- Respect source restrictions and rate limits.

Output

- First raw dataset
 - Collection log
 - Unit tests
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Day 6 — Second Source Collector

- Add a second data source.
- Follow the same collector interface.
- Compare available fields with the first source.
- Store source-specific fields without forcing premature standardization.

Output

- Two-source raw dataset
 - Collector coverage comparison
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Day 7 — Raw Data Quality Review

Analyze:

- Record counts
- Missing fields
- Duplicate listings
- Invalid dates
- Price availability
- Age-group availability
- Historical depth
- Sold-out-status availability

Output

- Pilot raw-data quality report
 - Decision on whether the sources are sufficient
-

Day 8 — Cleaning Pipeline

Implement standardization for:

- City
- Locality
- Workshop category
- Age group
- Price
- Duration
- Delivery format
- Event type
- Status

Output

- Standardized event table
 - Cleaning tests
 - Invalid-record report
-

Day 9 — Duplicate and Entity Resolution

Build logic for:

- Duplicate event listings
- Recurring sessions
- Organizer-name variations
- Venue-name variations
- Events appearing across multiple platforms

Output

- Deduplicated event dataset
 - Organizer master table
 - Manual-validation sample
-

Day 10 — Core Database Model

Create:

- Organizer dimension
- Category dimension
- Audience dimension
- Venue dimension
- Date dimension
- Event-listing fact
- Event-occurrence fact
- Price snapshot
- Availability snapshot

Output

- Core PostgreSQL model
 - Table-grain documentation
 - Database diagram
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Day 11 — dbt Setup

- Configure dbt-postgres.
- Define source tables.
- Build staging models.
- Add uniqueness and not-null tests.
- Generate dbt documentation.

Output

- Working dbt project
 - Passing dbt tests
 - Initial lineage graph
-

Day 12 — Analytics Marts

Create marts for:

- Market overview
- City comparison
- Category demand
- Audience demand
- Pricing
- Organizer performance
- Event status
- Power BI consumption

Output

- Analysis-ready SQL models
 - Stable Power BI tables
-

Day 13 — Exploratory Analysis

Analyze:

- Events by city
- Events by category
- Pricing distribution
- Age groups
- Event formats
- Organizer activity
- Missing-data patterns

Output

- EDA notebook
 - Initial observations
 - Data limitations
-

Day 14 — Demand Metrics

Create:

- Event-frequency metric
- Repeat-event rate
- Sold-out rate
- Review-volume metric
- Premium-price share
- Discount dependency
- Recurring-program availability
- Demand-signal coverage

Output

- Metric-definition document
 - Metric-validation notebook
-

Day 15 — Pilot Review Gate

Decide:

- Which sources should be retained
- Which collectors require improvement
- Whether Hyderabad and Bengaluru are comparable
- Whether the model can scale to four additional metros
- Which data gaps need alternative proxies

Output

- Pilot findings
 - Go, revise, or stop decision
 - Revised expansion plan
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Day 16 — Expand to Mumbai

- Reuse the validated pipeline.

- Add city-specific sources only when necessary.
- Validate localities and pricing.

Day 17 — Expand to Delhi NCR

- Support Delhi, Gurugram, and Noida.
- Preserve city and metro-region distinctions.

Day 18 — Expand to Pune

- Run existing collectors.
- Validate local organizers and category coverage.

Day 19 — Expand to Chennai

- Run existing collectors.
- Review language and locality variations.

Days 16–19 outputs

- Six-city standardized dataset
- Source coverage report
- City-level quality checks

Day 20 — Historical and Trend Analysis

Analyze maximum recoverable history:

- Events by year
- Category growth
- City growth
- New organizer entry
- Organizer longevity
- Online versus in-person change
- Seasonal patterns

Output

- Historical trend notebook
 - Coverage-adjusted trend findings
-

Day 21 — Hypothesis Testing

Test the strongest supported hypotheses, such as:

- Weekend versus weekday demand
- Premium pricing versus sold-out status
- Children's events versus adult events
- Camps versus individual workshops
- Dedicated studios versus mobile formats
- Review activity versus event frequency

Output

- Hypothesis-testing notebook
 - Effect sizes
 - Limitations and interpretation
-

Day 22 — Business Feasibility

Create low, base, and high scenarios for:

- Dedicated studio
- Home-based model
- Rented venue
- Mobile workshops
- Online or hybrid model

Output

- Startup-cost assumptions
 - Revenue-per-event model
 - Break-even attendance
 - Monthly booking requirements
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Day 23 — Power BI Data Model

- Connect Power BI to PostgreSQL.
- Import Power BI marts.
- Define relationships.
- Create core DAX measures.
- Validate totals against SQL.

Output

- Working Power BI semantic model

- Reconciliation checklist
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Day 24 — Dashboard Development

Build dashboard pages for:

- Market overview
- City comparison
- Category demand
- Audience analysis
- Pricing
- Seasonality
- Business-model comparison
- Opportunity ranking

Output

- Power BI dashboard
 - Dashboard QA results
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Day 25 — Final Findings and Portfolio Package

Produce:

- Executive summary
- Methodology
- Market findings
- Hypothesis results
- Business opportunity ranking
- Recommended pilot business
- Risks and limitations
- GitHub README
- Portfolio explanation
- Presentation outline

Output

- Completed repository
 - Dashboard
 - Final analytical report
 - Portfolio-ready case study
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Daily Synchronization Process

At the end of every Codex work session, capture:

```
Day:
Objective completed:
Files created or modified:
Commands executed:
Tests passed:
Tests failed:
Data collected:
Important assumptions:
Problems or blockers:
Git commit:
Next recommended action:
```

Codex should update `docs/project_log.md` with this information.

Share one of the following with ChatGPT after each session:

- Codex summary
- Changed files
- `git diff --stat`
- Test output
- Error output
- Relevant code files
- Sample data
- Notebook findings

ChatGPT will then:

1. Compare the result with the project plan.
2. Review technical and analytical correctness.
3. Identify missing work.
4. Explain what the code is doing.
5. Prepare the next bounded Codex instruction.
6. Keep the business analysis aligned with the original objective.

Current Project Status

Completed

- Business objective defined

- Business questions defined
- High-level project plan defined
- GitHub repository created
- Codex initialized
- Proposed architecture created
- Pilot cities revised to Bengaluru and Hyderabad

Not started

- Repository implementation review
- Architecture update
- Taxonomy
- Source register
- PostgreSQL setup
- Data collection

Next Step

Day 1: Repository and architecture review

Do not begin full data collection yet. First, inspect what Codex created, update the pilot scope, run tests, and approve the repository structure.